Institute of Computer Science
University of Augsburg

Bachelor's Thesis

Title of the Bachelor's or Master's Thesis

Name

Student ID Number: Student ID Number
First Reviewer: Prof. Dr. Jakob Siegfried Kottmann
Second Reviewer: Prof. Dr. Mónica Benito
Scientific Supervisor:
Date: June 15, 2026

written at
Chair of Quantum Algorithms
Prof. Dr. Jakob Siegfried Kottmann
Institut of Computer Science
University of Augsburg
D–86135 Augsburg, Germany

Abstract

Describe the content of the present thesis here. Briefly explain the starting point of the thesis and outline its objectives. Then, discuss the methods used and the results achieved.

Contents

1. Introduction

2. Theoretical Fundamentals

2.1. Fundamentals of Quantum Information

This section outlines the core principles of quantum information theory, establishing the essential theoretical vocabulary necessary for navigating the complexities of quantum computation.

We begin by defining the mathematical representation of quantum states. To describe composite systems consisting of multiple interacting components, we introduce the tensor product, which enables us also to represent the superpostions. Within this combined space, measure physical properties and defining interactions requires linear operators and observables. Finally, we detail the unitary operations and their underlying mathematical generators. Together, these foundational concepts provide the prerequisite knowledge required to understand the quantum formalisms discussed in subsequent chapters.

2.1.1. Quantum States

In an n-dimensional Hilbert space with a chosen orthonormal basis, a quantum state, denoted by the ket $|\psi\rangle$, can be represented as a n-dimensional complex column vector $\vec{\psi}$:

$$|\psi\rangle \doteq \vec{\psi} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}. \tag{2.1}$$

Its corresponding dual state, denoted by the bra $\langle\psi|$, is represented by the conjugate transpose of this column vector, resulting in an n-dimensional row vector $\vec{\psi}^\dagger$:

$$\langle\psi| \doteq \vec{\psi}^\dagger = \begin{pmatrix} c_1^* & c_2^* & \dots & c_n^* \end{pmatrix}. \tag{2.2}$$

To ensure consistency with the probability interpretation, quantum states must be normalized to unity. Therefore, for a quantum state $|\psi\rangle$, the sum of the absolute squares of its probability amplitudes must equal to one:

$$|c_1|^2 + |c_2|^2 + \cdots + |c_n|^2 = 1. \tag{2.3}$$

[3, chapter 1.1, 2.1] [5, chapter 1.2.1]

2.1.2. Tensor Products

In quantum mechanics, multi-particle systems are constructed within a composite Hilbert space using tensor products. The tensor product combines two or more distinct vector spaces into a single, larger space. Mathematically, if we consider a vector space of dimension d_1 and a second vector space of dimension d_2, the resulting composite space has a dimension of $d_1 \cdot d_2$ (i.e., $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2} \to \mathbb{C}^{d_1 \cdot d_2}$).

Constructing a multi-qubit basis state is achieved by taking the tensor product of the individual single-qubit states. For instance, if one qubit is in the state $|0\rangle$ and a second is in the state $|1\rangle$, the joint state of the two-qubit system is $|0\rangle \otimes |1\rangle$, which is conventionally abbreviated as $|01\rangle$. Because quantum systems can exist in superposition, a general composite state is expressed as a linear combination of all possible tensor product basis states. For a two-qubit system, this yields a four-dimensional complex vector:

$$|\psi\rangle = c_{00} |00\rangle + c_{01} |01\rangle + c_{10} |10\rangle + c_{11} |11\rangle . \tag{2.4}$$

The tensor product formalism also provides a definition of entanglement. A multi-particle state is defined as entangled if it cannot be factored into a simple tensor product of individual, single-particle states (i.e., $|\psi\rangle \neq |\phi\rangle \otimes |\varphi\rangle$).

When quantum operators act on these composite Hilbert spaces, their tensor products obey several fundamental algebraic properties:

- **Non-commutativity:** The order of the tensor product strictly matters; in general, $X \otimes Y \neq Y \otimes X$.
- **Bilinearity:** The operation distributes over addition, such that $X \otimes (Y + Z) = X \otimes Y + X \otimes Z$.
- **Mixed-Product Property:** Operators act independently on their respective subsystems, following the rule $(A \otimes B)(C \otimes D) = AC \otimes BD$.
- **Eigensystems:** If an operator is formed by the tensor product of two matrices, its eigenvectors are the tensor products of the individual eigenvectors ($|a_k\rangle \otimes |b_l\rangle$), and the corresponding eigenvalues are the products of the individual eigenvalues ($a_k \cdot b_l$).

[3, chapter 2.3, A.8]

2.1.3. Operators and Observables

A linear operator acts on a ket from the left to produce another ket vector, or on a bra from the right to produce another bra vector:

$$X |\alpha\rangle = |\beta\rangle , \tag{2.5}$$

$$\langle\alpha| X = \langle\beta| . \tag{2.6}$$

An operator X is defined as the **null operator** if, for any arbitrary ket $|\alpha\rangle$, it satisfies:

$$X |\alpha\rangle = 0. \tag{2.7}$$

Operators can be added ($X + Y = Y + X$) and multiplied. While operator multiplication is generally noncommutative ($XY \neq YX$), it is strictly associative:

$$X(YZ) = (XY)Z = XYZ. \tag{2.8}$$

Because quantum states are represented as vectors in $\mathbb{C}^n$, a linear operator can be represented by a complex $n \times n$ matrix U. Given an input ket $|\psi\rangle$, the resulting state $|\phi\rangle$ is calculated via matrix-vector multiplication:

$$|\phi\rangle = U |\psi\rangle . \tag{2.9}$$

Crucially, physical operations that evolve a quantum state must preserve its probability normalization. Consequently, physical evolution operators are restricted to unitary matrices, ensuring that the inner product—and thus the normalization—remains invariant

An **observable** represents a physical quantity of a system that can be experimentally measured. Classical examples include position, momentum, angular momentum, and energy. In an N-qubit system, an observable is defined mathematically as a $2^N \times 2^N$ Hermitian matrix H. Because the operator is Hermitian, it guarantees that all possible measurement outcomes are strictly real numbers and correspond directly to the eigenvalues of H. Crucially, the act of measurement fundamentally alters the system: upon obtaining a specific eigenvalue, the quantum state instantaneously collapses into the corresponding eigenstate of that observable

Due to the inherent probabilistic nature of quantum mechanics, we generally cannot predict the outcome of a single measurement with absolute certainty. Instead, we evaluate the **expectation value**, which represents the statistical average of measured values over a large ensemble of identically prepared systems

In the context of this thesis, the primary observables of interest are Hamiltonians, which represent the total energy of the physical system. Let H be the Hamiltonian.

We conventionally assume that its resulting energy eigenvalues are ordered from the ground state upwards ($E_0 \leq E_1 \leq E_2 \leq \dots$), satisfying the eigenvalue equation:

$$H\left|E_i\right\rangle = E_i\left|E_i\right\rangle . \tag{2.10}$$

Given a hamiltonian Operator H and a quantum state $|\psi\rangle$ the expectation value can be calculated as:

$$\langle H\rangle_\psi := \langle\psi| H |\psi\rangle . \tag{2.11}$$

[3, chapter 2.2, 5.3] [5, chapter 1.2, 1.4.1] [6, chapter 1.1.1]

2.1.4. Unitary Operations and Generators

The physical transformations applied to quantum states are represented by unitary operators. To construct parameterized quantum gates, these unitary operations are typically expressed via the exponential map:

$$U(\theta) = e^{-i\frac{\theta}{2}G}, \tag{2.12}$$

where θ is a real-valued scalar parameter (such as a time or rotation angle), and G is a Hermitian matrix ($G = G^\dagger$) known as the generator of the operation. [2, chapter 1.2.12] [4, chapter II.]

2.2. Fermionic Many-Body Systems

Fermions are fundamental particles such as electrons, protons, and neutrons, constitute the primary building blocks of matter. In the context of computational many-body physics, the challenge of simulating these systems lies not in their isolated particle properties, but in the collective quantum mechanical rules governing their many-body dynamics. Specifically, a simulation must describe the rules of fermions and their interactions accurately. [2, chapter 6.1]

2.2.1. Indistinguishability

In classical physics, identical objects can be tracked along distinct, observable trajectories. In quantum mechanics, however, identical particles are fundamentally indistinguishable; it is physically impossible to track their individual paths or assign them persistent labels without collapsing or disturbing the system's state. Because measurable physical quantities are calculated as expectation values, which rely on the product of the wavefunction and its complex conjugate ($\langle\psi| H |\psi\rangle = \sum_i |c_i|^2 E_i$), the indistinguishability principle dictates that these observables must remain strictly invariant under the permutation (exchange) of any two identical particles. [5, chapter 7.1, 7.2] [2, chapter 6.1]

2.2.2. Wavefunction Symmetry and Antisymmetry

In many-body quantum mechanics, systems of identical particles are tightly constrained by symmetry. The total wavefunction of such a system must be either entirely symmetric or entirely antisymmetric under the exchange of any two identical particles.

Antisymmetric Wavefunctions: Fermions are strictly governed by totally antisymmetric wavefunctions. Consequently, exchanging the spatial and spin coordinates of any two identical fermions introduces a global phase factor of -1 to the wavefunction. The Pauli Exclusion Principle is a direct mathematical corollary of this antisymmetry; if two identical fermions were to occupy the identical quantum state, their exchange would require the wavefunction to equal its own negative, implying the wavefunction must identically vanish.

Symmetric Wavefunctions: Bosons, conversely, are described by totally symmetric wavefunctions. Exchanging any two identical bosons yields a phase factor of $+1$, leaving the wavefunction fundamentally unaltered. Free from the antisymmetric cancellation constraints that restrict fermions, identical bosons can condense into the same quantum state.

[5, chapter 7.1, 7.2] [2, chapter 6.1]

2.2.3. Pauli Exclusion Principle

The Pauli Exclusion Principle, formulated by Wolfgang Pauli in 1925, is a direct physical consequence of the indistinguishability applied specifically to fermions. It establishes that no two identical fermions can simultaneously occupy the exact same quantum state within a given system. Originally conceptualized to explain atomic electron configurations, where no two electrons can share the same four quantum numbers, this principle is a defining, non-negotiable characteristic of fermionic behavior. This exclusion must be preserved. Any application of creation ($a_i^\dagger$) or annihilation (a_i) operators must naturally prevent multiple occupancies. [5, chapter 7.1, 7.2]

2.3. The Second Quantization Framework

In the first quantization formalism, describing a many-body fermionic system requires the explicit construction of fully antisymmetric wavefunctions. As the number of particles increases, tracking the coordinates of individual indistinguishable particles becomes hard to deal. The second quantization framework fundamentally resolves this bottleneck. Rather than defining explicit wavefunctions for individual particles, it shifts the perspective to the occupation of predefined single-particle states within a Fock space. Particle indistinguishability and wavefunction antisymmetry are not discarded, they are rather embedded directly into the anticommutation algebra of the fermionic creation and annihilation operators. This provides a scalable, operator-driven foundation.

[6, chapter 2.1] [2, chapter 6.5] [5, chapter 7.7.1]

2.3.1. Fock Space

To mathematically formalize the second quantization framework, we require a state space capable of accommodating a variable number of particles. This is achieved by constructing the Fock space, $\mathcal{F}$, which is defined as the direct sum of N-particle Hilbert spaces. This structure allows us to express states without being restricted to a fixed particle number.

Because this space utilizes the second quantization formalism, it does not track the explicit spatial coordinates of each particle. Instead, states in the Fock space are expressed using the occupation number representation:

$$|\psi\rangle = |n_1, n_2, \ldots, n_i, \ldots\rangle \, . \tag{2.13}$$

In this notation, each index i corresponds to a specific single-particle state (or spin-orbital), and n_i denotes its respective occupation number. Note that due to the Pauli exclusion principle, the occupation number for fermions is strictly restricted to $n_i \in \{0, 1\}$.

To illustrate this representation, we can define several fundamental states. The vacuum state, representing a system empty system without any particles, is denoted as:

$$|0, 0, \ldots, 0, \ldots\rangle = |\mathbf{0}\rangle \, . \tag{2.14}$$

A single-particle state, where only the i-th spin-orbital is occupied, is written by setting $n_i = 1$:

$$|0, \ldots, n_i = 1, \ldots\rangle = |k_i\rangle \, . \tag{2.15}$$

More complex, multi-particle states are systematically generated by applying creation and annihilation operators to these basis states. For example in a system with 4 spin-orbitals, the occupation vector $|0110\rangle$ indicates that the second and third spin-orbitals are occupied by particles, while the first and fourth remain empty.

2.3.2. Fermionc Operators

In quantum mechanics, a many-body system of identical particles is described using the formalism of second quantization. Within this framework, the physical state of a system is manipulated via fundamental mathematical tools known as creation ($\hat{a}_k^\dagger$) and annihilation ($\hat{a}_k$) operators. These operators act directly upon a quantum state: $\hat{a}_k^\dagger$ creates a fermion in a specific spin-orbital k, while $\hat{a}_k$ destroys a fermion in that same orbital. Because fermions inherently obey the Pauli Exclusion Principle, these operators naturally enforce strict physical boundaries; applying a creation operator to an already occupied orbital, or an annihilation operator to an empty one, identically annihilates the wavefunction (yielding zero). These characteristics can be noted down as:

$$\hat{a}|0\rangle = 0, \tag{2.16}$$

$$\hat{a}^\dagger|0\rangle = |1\rangle, \tag{2.17}$$

$$\hat{a}|1\rangle = |0\rangle, \tag{2.18}$$

$$\hat{a}^\dagger|1\rangle = 0. \tag{2.19}$$

[6, chapter 2.1]

2.4. Mapping Fermions to Quantum Hardware

The objective of a fermionic simulator is to evaluate the dynamics of many-body fermionic systems using qubit-based architectures. However, quantum hardware exclusively executes operations on distinguishable qubits. Consequently, a mathematically encoding scheme is required to translate fermionic behavior into a native qubit representation. This section first examines the fundamental algebraic mismatch between fermions and qubits, and subsequently introduces the specific mapping mechanisms to bridge this hardware gap.

2.4.1. The Hardware Constraint: Fermions vs. Qubits

While fermions are physical particles governed by indistinguishability, qubits are idealized, fully distinguishable two-state systems that serve as the fundamental units of quantum computation. In a quantum processor or qubit-based simulator, individual qubits are independently addressable. This distinguishability yields a crucial algebraic divergence: operators acting on disparate qubits naturally commute:

$$[A, B] = AB - BA = 0 \quad \Rightarrow \quad AB = BA \tag{2.20}$$

In contrast, fermionic creation ($\hat{a}_i^\dagger$) and annihilation ($\hat{a}_i$) operators must anticommute:

$$\{\hat{a}_i^\dagger, \hat{a}_j^\dagger\} = \hat{a}_i^\dagger\hat{a}_j^\dagger + \hat{a}_j^\dagger\hat{a}_i^\dagger = 0 \quad \Rightarrow \quad \hat{a}_i^\dagger\hat{a}_j^\dagger = -\hat{a}_j^\dagger\hat{a}_i^\dagger \tag{2.21}$$

This chrectersitc is needed to preserve wavefunction antisymmetry and enforce the Pauli Exclusion Principle.

[5, chapter 3.11.1, 7.1, 7.7] [3, chapter 1.]

2.4.2. The Jordan-Wigner Transformation

To simulate fermionic systems on quantum hardware, an encoding scheme is required to map the physical particles to the computational framework. The Jordan-Wigner (JW) transformation serves as the foundational mechanism for this mapping, establishing a direct bijection between N fermionic spin-orbitals and N individual qubits. Within this framework, the computational basis states directly represent the Fock space occupation number vectors: a qubit in state $|1\rangle$ denotes an occupied spin-orbital, while $|0\rangle$ denotes an empty one.

However, preserving the indistinguishability of fermions presents a significant hardware constraint. Fermionic creation and annihilation operators are mathematically constrained to strictly anticommute to guarantee that the many-particle wavefunction remains totally antisymmetric under particle exchange. Because quantum hardware utilizes distinguishable qubits whose operators naturally commute, fermionic operators cannot be mapped to single qubits in a simple one-to-one fashion. Instead, they must be explicitly constructed from composite qubit operators.

Under the standard Jordan-Wigner orbital-to-qubit mapping, the fermionic operators are defined as:

$$\hat{a}_k^\dagger = \mathbb{I}^{\otimes(k-1)} \otimes \hat{\sigma}_k^+ \otimes \hat{\sigma}_z^{\otimes(N-k)}, \tag{2.22}$$

$$\hat{a}_k = \mathbb{I}^{\otimes(k-1)} \otimes \hat{\sigma}_k^- \otimes \hat{\sigma}_z^{\otimes(N-k)}. \tag{2.23}$$

This composite definition relies on two functional components to artificially replicate fermionic behavior within an inherently commuting qubit environment:

Ladder Operators ($\hat{\sigma}_k^\pm$): Defined using the standard Pauli matrices as:

$$\hat{\sigma}_k^\pm = \frac{1}{2}(\hat{\sigma}_x \pm i\hat{\sigma}_y). \tag{2.24}$$

These operators execute the local "creation" or "annihilation" at the specific target qubit k by raising or lowering the computational basis state between $|0\rangle$ (empty) and $|1\rangle$ (occupied).

Parity Strings ($\hat{\sigma}_z^{\otimes(N-k)}$): A non-local tensor product of Pauli-Z operators acting on the adjacent qubits. This string encodes the necessary parity information into the phase of the wavefunction.

When evaluated against a quantum state, these non-local parity strings ($\hat{\sigma}_z$) systematically scan all qubits succeeding (or preceding, depending on the chosen convention) the target site k. Because the Pauli-Z operator acts as the identity on $|0\rangle$ and applies a phase of -1 to $|1\rangle$, this string iteratively multiplies a global phase factor of -1 for every occupied state it encounters. This mechanism systematically and globally replicates the exact antisymmetric phase inversions demanded by the Pauli Exclusion Principle.

[6, chapter 2.1] [1, chapter III.B] [4, chapter II.E] [3, chapter 1.1, 5.3]

2.5. Simulating Excitations

In quantum mechanics, an excitation describes the transition of a particle, such as an electron, from a lower-energy orbital to a higher-energy state. Within the context of computational many-body physics, these phenomena can manipulate the quantum state of the system.

We saw before that fermions differ from qubits, also in the defintion of the excitation of the both particle groups differences can be seen. In both cases, excitations are enacted via unitary evolution operators; however, the explicit construction of the underlying mathematical generators must be changed to obey the respective commutation or anticommutation rules of the system. This is detailed in this section.

2.5.1. Qubit Excitations

To dynamically evolve a quantum state on hardware, the most direct approach is to apply excitations directly within the computational qubit space. These operations, commonly referred to as qubit excitations or orbital rotations, move particle occupations between sites. Instead of utilizing true fermionic creation ($\hat{a}^\dagger$) and annihilation ($\hat{a}$) operators, qubit excitations are constructed natively from the Ladder Operators.

Mathematically, a single-qubit excitation describing a transition between an occupied qubit i and a target qubit b is defined by the unitary operator:

$$U_i^b(\theta) = e^{\theta\left(\hat{\sigma}_b^+\hat{\sigma}_i^- - \hat{\sigma}_i^+\hat{\sigma}_b^-\right)}. \tag{2.25}$$

This can be equivalently expressed in terms of standard Pauli matrices (absorbing the scalar factors into the rotation parameter θ) as (see appendix for algebraically derivation):

$$U_i^b(\theta) = e^{-i\theta\left(\hat{\sigma}_x^b\hat{\sigma}_y^i - \hat{\sigma}_y^b\hat{\sigma}_x^i\right)}. \tag{2.26}$$

Note that $\hat{\sigma}_x$ and $\hat{\sigma}_y$ refers to the Pauli-X and Pauli-Y operators. This unitary operator acts locally on the target qubits, functioning as a rotation matrix exclusively within the $\{|01\rangle, |10\rangle\}$ subspace to swap the occupation states.

[1, Chapter IV.A.1.b.]

This concept naturally extends to higher-order excitations. Given a set of target orbitals denoted by the vector $\boldsymbol{k}$, and a set of occupied orbitals denoted by $\boldsymbol{l}$, the multi-qubit excitation generator is constructed directly from products of the corresponding raising and lowering operators:

$$G_{\boldsymbol{l}}^{\boldsymbol{k}} = i\left(\prod_{n} \hat{\sigma}_{k_n}^{-} \prod_{m} \hat{\sigma}_{l_m}^{+} - \text{h.c.}\right). \tag{2.27}$$

2.5.2. Fermionic Excitations

While qubit excitations are highly hardware-efficient and successfully redistribute particle occupations, they lack a crucial physical property: they do not natively track the non-local antisymmetric phase shifts required when identical fermions exchange positions. To fully capture the physical reality of a many-body system and enforce the Pauli Exclusion Principle globally, we must utilize true fermionic excitations.

Fermionic excitations are represented by a generator G, constructed strictly from the anticommuting creation ($\hat{a}^\dagger$) and annihilation ($\hat{a}$) operators introduced during second quantization. For instance, the generator G_{pq} describes a single fermionic excitation, denoting the transition of an electron from the p-th spin-orbital to the q-th spin-orbital:

$$G_{pq} = i(\hat{a}_p^\dagger \hat{a}_q - \hat{a}_q^\dagger \hat{a}_p). \tag{2.28}$$

Higher-order excitations are constructed in a similar mathematical manner. The generator G_{pqrs} models a double excitation, accounting for the simultaneous transition of one electron between the p-th and q-th orbitals, and a second electron between the r-th and s-th orbitals:

$$G_{pqrs} = i(\hat{a}_p^\dagger \hat{a}_q \hat{a}_r^\dagger \hat{a}_s - \text{h.c.}). \tag{2.29}$$

Here, h.c. denotes the Hermitian conjugate of the preceding term. This standard notation simplifies the equation while ensuring the operator correctly captures the reverse electron exchange process to maintain Hermiticity, analogous to the subtracted term in the single excitation.

Generalizing this concept, an n-fold fermionic excitation can be compactly represented by a generator $G_{\mathbf{pq}}$, which encapsulates all possible simultaneous electron exchanges between a set of occupied orbitals $\{p_k\}$ and a set of target orbitals $\{q_k\}$. Because this represents n simultaneous events, it is defined as a product of n operator pairs:

$$G_{\mathbf{pq}} = i\left(\prod_{k=1}^{n} \hat{a}_{p_k}^\dagger \hat{a}_{q_k} - \text{h.c.}\right). \tag{2.30}$$

To apply these generators within a quantum simulator or on physical quantum hardware, they are mapped via the Jordan-Wigner transformation and exponentiated into a unitary operation (e.g., $U = \exp(-i\theta G_{\mathbf{pq}})$).

[4, chapter II.][1, chapter III.A]

3. Algorithmic Design of the Simulator

3.1. Matrix-Free Direct State Evolution

The core computational challenge in simulating many-body quantum systems lies in the efficient application of excitation operators to the underlying state vector. Constructing and applying global unitary matrices is highly memory-prohibitive for large Hilbert spaces. Therefore, the simulator leverages a direct state-vector approach, exploiting the algebraic properties of the excitation generator to execute updates directly on the computational basis.

To translate this theoretical framework into a high-performance algorithmic routine, we must deconstruct the evaluation of the unitary operation defined in Equation (2.12). As an initial step, we isolate the fundamental mechanics by considering the qubit excitation framework. This isolated model provides a computational baseline that is then subsequently generalized to incorporate fermionic anticommutation relations.

3.1.1. Qubit Baseline

Rather than relying on computationally prohibitive full matrix exponentiation to evaluate the time-evolution operator $U = \exp\left(\sum \theta \hat{E}\right)$, the direct state evolution approach exploits the algebraic properties of the generator to execute localized state updates. To compute the excitation efficiently, we must simplify the evaluation of the unitary operation defined in Equation (2.12). As an initial simplification, we consider the qubit excitation framework, which can later be generalized to account for fermionic differences.

Consider an initial quantum state upon which the excitation is applied. Recalling the generalized generator from Equation (2.27), the operation is fully defined by the orbital sets $\boldsymbol{k}$ and $\boldsymbol{l}$, expressing which electrons are exchanged. With these target orbitals and the initial state, the algorithm possesses all the necessary parameters. Given the generator $G_{\boldsymbol{l}}^{\boldsymbol{k}}$, the unitary operator can be expanded as:

$$U(\theta) = e^{-i\frac{\theta}{2}G_{\boldsymbol{l}}^{\boldsymbol{k}}} = \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) G_{\boldsymbol{l}}^{\boldsymbol{k}} + \left(1 - \cos\left(\frac{\theta}{2}\right)\right) P_0. \tag{3.1}$$

Here, P_0 represents the projector onto the nullspace of the generator $G_{\boldsymbol{l}}^{\boldsymbol{k}}$. This nullspace is spanned by all computational basis states $|\boldsymbol{i}\rangle$ for which $G_{\boldsymbol{l}}^{\boldsymbol{k}}|\boldsymbol{i}\rangle = 0$.

When applied to a specific electronic configuration $|\Phi\rangle$, the unitary transformation acts trivially (as the identity operator) if the configuration resides within the nullspace of the generator. Otherwise, it induces a rotation between the original configuration and the n-fold excited configuration:

$$U(\theta)\,|\Phi\rangle = \begin{cases} |\Phi\rangle\,, & \text{if } P_0\,|\Phi\rangle = |\Phi\rangle \\ \left(\cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) G\right)|\Phi\rangle\,, & \text{else.} \end{cases} \tag{3.2}$$

To implement the active transformation algorithmically, we examine the two components of the rotational term. The first term, governed by $\cos\left(\frac{\theta}{2}\right) I$, is computationally straightforward. It is expressed as a direct scaling of the initial state's probability amplitude, which can be expressed algorithmically as an update:

$$c_{\boldsymbol{i}} \leftarrow c_{\boldsymbol{i}} \cos\left(\frac{\theta}{2}\right). \tag{3.3}$$

The second component of the rotation governs the coupling and is more complex as it involves the explicit action of the generator. The term $-i\sin\left(\frac{\theta}{2}\right) G_{\boldsymbol{l}}^{\boldsymbol{k}}$ can be decomposed into two distinct operational steps. First, the algorithm computes the partner state $|\boldsymbol{i'}\rangle$ resulting from the generator's application onto the basis state:

$$c\,|\boldsymbol{i'}\rangle = -iG_{\boldsymbol{l}}^{\boldsymbol{k}}\,|\boldsymbol{i}\rangle\,, \tag{3.4}$$

$$-iG_{\boldsymbol{l}}^{\boldsymbol{k}} = \left(\prod_n \sigma_{k_n}^{-}\sigma_{l_n}^{+} - \text{h.c.}\right). \tag{3.5}$$

Here, the partner state $|\boldsymbol{i'}\rangle$ represents the specific electronic configuration generated by the excitation, which is also introduced into the superposition. By absorbing the factor of $-i$ into the generator's evaluation, the transitional coefficient c is restricted to purely real values and even $c = \pm 1$. The sign of c depends strictly on the directionality of the particle exchange (i.e., whether the operation acts as an excitation or de-excitation on the specific state).

Finally, this term induces an amplitude update on the newly acquired partner state. The corresponding probability amplitude is updated as:

$$c_{\boldsymbol{i'}} \leftarrow c_{\boldsymbol{i'}} + c\sin\left(\frac{\theta}{2}\right). \tag{3.6}$$

[4, Chapter II.A]

3.1.2. Incorporating the Fermionic Phase

While the foundational baseline algorithm effectively handles qubit-based rotations, it is inherently insufficient for simulating the characteristics of true fermionic many-body systems. To accurately model fermions, the computational routine must replace the qubit excitation generator with its fermionic counterpart, the multi-excitation generator $G_{\boldsymbol{pq}}$, as defined in Equation (2.30).

As established in Section 2.4.2, fermionic creation and annihilation operators can be mapped to qubit representations via the Jordan-Wigner transformation. This mapping explicitly reveals the computational difference between the two algorithmic frameworks. While isolated qubit excitations can be described using standard ladder operators ($\hat{\sigma}^{\pm}$), fermionic operators require the additional application of a parity string ($\hat{\sigma}_z^{\otimes(N-k)}$). This parity string is a tensor product of Pauli-Z operators spanning all intermediate qubits located between the target orbitals of the excitation.

When acting upon the computational basis, these intermediate Pauli-Z operators return an eigenvalue of $+1$ if an orbital is empty ($|0\rangle$) and -1 if it is occupied ($|1\rangle$). Consequently, the parity string algorithmically counts the number of occupied orbitals located between the origin and destination of the electron jump. If the intermediate occupation is even, the -1 eigenvalues perfectly cancel, multiplying the overall state by a global $+1$ (resulting in no phase shift). Conversely, if the intermediate occupation is odd, the -1 eigenvalues do not cancel, and a net -1 phase is applied to the quantum state amplitude.

This fundamental difference in dynamic behavior originates from the Pauli exclusion principle and the strict antisymmetry required of fermionic wavefunctions. Algorithmically, it is crucial to note that while the nullspace projector P_0 acts purely based on the occupancy of the initial and target orbitals and ignores phase dynamics complety. Therefore the correct evaluation of the active state rotation relies entirely on accurately tracking and applying this parity phase.

Note: The switch from qubit to fermionic on quantum Hardware is because of the Parity String very complex. To apply the long chain of Pauli-Z operators the quantum circuit requires many deep CNOT "ladders´´. (read more about it in [1, chapter III.B])

[1, chapter III.B] [6, chapter 2.1.1]

3.1.3. Bitwise Representation and Algorithmic Mapping

To transition from the theoretical framework to a high-performance computational model, the continuous physics of quantum operators must be mapped onto discrete, logical operations. Before discussing the specific software architecture, the matrix-free evolution can be abstracted into four core algorithmic steps.

State Representation and Bitwise Excitations. The computational basis states are mapped to integer bitstrings, where each bit represents a spin-orbital (0 denoting an empty state, 1 denoting an occupied state). The geometric application of an excitation operator corresponds directly to flipping the occupation states at the targeted origin ($\boldsymbol{k}$) and destination ($\boldsymbol{l}$) indices. Algorithmically, this state transition is executed using a bitwise exclusive OR (XOR, $\oplus$) operation, allowing for instantaneous configuration updates without the overhead of algebraic matrix multiplication.

Evaluation of the Nullspace Projector (P_0). Instead of constructing a dense projector matrix, the nullspace is evaluated via a highly efficient boolean verification loop. For an excitation generator to act non-trivially on a state, the electron occupancy across all origin orbitals must be uniform (either completely full or completely empty), and the destination orbitals must hold the exact inverse configuration. If these logical conditions are not met, the state resides in the nullspace and the algorithm trivially proceeds to the next basis configuration.

Parity Evaluation via Bitmasks. To enforce fermionic antisymmetry, the algorithm must dynamically track the Jordan-Wigner parity phase. The tensor product of Pauli-Z operators is translated into a bitmask operation that isolates all orbitals topologically located between the target indices. A population count then calculates the number of active fermions within this isolated subspace. Crucially, this parity evaluation is applied sequentially for the annihilation and creation steps. This dynamic separation is necessary to respect the non-commutative ordering of the fermionic operators (i.e., applying T versus $T^\dagger$), ensuring the correct cumulative parity phase (± 1) is tracked.

Sparse Delta-Update Strategy. The fundamental advantage of this direct state-vector approach is its sparse memory profile. Rather than iterating over the entire Hilbert space to apply the identity operator or trigonometric scaling, the algorithm utilizes a delta-update strategy. For actively coupled states, the probability amplitude is modified in place by adding the difference $c_i(\cos(\theta/2) - 1)$, which effectively evaluates to the target scaling $c_i \cos(\theta/2)$. Simultaneously, the geometrically determined partner state is injected into the superposition with the newly calculated amplitude $\pm\text{phase} \cdot c_i \sin(\theta/2)$, entirely bypassing zero-element computations.

4. Implementation in C++

- Explanation
- Decision on implementation
- Software architecture
- Effizienz strategy:
  - Why C++ (Memory management, Speed, etc.)
  - Data structures
  - Optimization of the loop

5. Evaluation and results

- Check for Correctness with Tequila
- Performance-analyse: how does the computation scale with difference states
- Comparison of different approaches

6. Summary and Outlook

List of Figures

List of Tables

Bibliography

[1] A. Anand, P. Schleich, S. Alperin-Lea, P. W. K. Jensen, S. Sim, M. Díaz-Tinoco, J. S. Kottmann, M. Degroote, A. F. Izmaylov, and A. Aspuru-Guzik. A quantum computing view on unitary coupled cluster theory, 2022.

[2] M. Benito and J. S. Kottmann. Quantum computing: Lecture notes ws25/26. Lecture Notes, University of Augsburg, 2025. Wintersemester 2025/26.

[3] J. S. Kottmann. Foundations of quantum information processing. Lecture Notes, University of Augsburg, 2025. Summer Semester 2025.

[4] J. S. Kottmann, A. Anand, and A. Aspuru-Guzik. A feasible approach for automatically differentiable unitary coupled-cluster on quantum computers, 2020.

[5] J. J. Sakurai and J. Napolitano. *Modern Quantum Mechanics*. Cambridge University Press, 3rd edition, 2020.

[6] P. Schleich, L. Mantilla Calderón, C. Sun, M. Bagherimehrab, A. Aldossary, J. S. Kottmann, and A. Aspuru-Guzik. *Quantum Computing for Quantum Chemistry*. ACS In Focus. American Chemical Society, 2025.

A. Appendix Chapter

A.1. Qubit Excitation Unitary Operation – Algebraic Derivation

Starting from the definition of the qubit excitation unitary in (2.25), we can express the Generatro in terms of the ladder operators defined in (2.24). To demonstrate this, we expand the constituent terms step-by-step. Expanding the first term, $(\hat{\sigma}_+^{(b)}\hat{\sigma}_-^{(i)})$, yields:

$$\frac{1}{4}\left(\hat{\sigma}_x^{(b)} + i\hat{\sigma}_y^{(b)}\right)\left(\hat{\sigma}_x^{(i)} - i\hat{\sigma}_y^{(i)}\right) = \frac{1}{4}\left(\hat{\sigma}_x^{(b)}\hat{\sigma}_x^{(i)} - i\hat{\sigma}_x^{(b)}\hat{\sigma}_y^{(i)} + i\hat{\sigma}_y^{(b)}\hat{\sigma}_x^{(i)} + \hat{\sigma}_y^{(b)}\hat{\sigma}_y^{(i)}\right). \tag{A.1}$$

Similarly, expanding the second term, $(\hat{\sigma}_-^{(b)}\hat{\sigma}_+^{(i)})$, gives:

$$\frac{1}{4}\left(\hat{\sigma}_x^{(b)} - i\hat{\sigma}_y^{(b)}\right)\left(\hat{\sigma}_x^{(i)} + i\hat{\sigma}_y^{(i)}\right) = \frac{1}{4}\left(\hat{\sigma}_x^{(b)}\hat{\sigma}_x^{(i)} + i\hat{\sigma}_x^{(b)}\hat{\sigma}_y^{(i)} - i\hat{\sigma}_y^{(b)}\hat{\sigma}_x^{(i)} + \hat{\sigma}_y^{(b)}\hat{\sigma}_y^{(i)}\right). \tag{A.2}$$

The definition of the generator requires subtracting these two expressions. Upon subtraction, the identical real components $\left(\hat{\sigma}_x^{(b)}\hat{\sigma}_x^{(i)}\text{and}\hat{\sigma}_y^{(b)}\hat{\sigma}_y^{(i)}\right)$cancel out entirely. Only the imaginary terms remain, reducing the final expression to:

$$\frac{1}{4}\left(-2i\hat{\sigma}_x^{(b)}\hat{\sigma}_y^{(i)} + 2i\hat{\sigma}_y^{(b)}\hat{\sigma}_x^{(i)}\right) = -\frac{i}{2}\left(\hat{\sigma}_x^{(b)}\hat{\sigma}_y^{(i)} - \hat{\sigma}_y^{(b)}\hat{\sigma}_x^{(i)}\right). \tag{A.3}$$

The $-\frac{1}{2}$ gets dropped for simplicity. [1, chapter IV.A]

A.2. bar

B. Second Appendix Chapter